David Krasowska

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Summary

David Krasowska is a Ph.D. candidate at Northwestern University, advised by Dr. Peter Dinda. His research began at Clemson University collaborating with Argonne National Laboratory on lossy compression for HPC scientific applications. He received the DOE Computational Science Graduate Fellowship for his graduate studies. Currently he explores raising the level of abstraction for programming emerging accelerators, such as multi-GPU systems and Processing-in-Memory.

Education

Ph.D. Candidate & M.S. in Computer Science, Northwestern University, Evanston, IL — 2023 – Now
B.S. in Computer Engineering, Clemson University, Clemson, SC — 2019 – 2022

Experience

Visiting Student, Los Alamos National Laboratory — Jun 2024 – Aug 2024

- Added support for UPMEM processing in memory (PIM) to the Legion runtime system.
- Evaluated Legion task-based programming models on PIM distributed clusters.
- Researched automated mapping of application kernels onto distributed heterogeneous systems.

Research Aide, Argonne National Laboratory — Jun 2022 – Jun 2023

- Expanded DRBSD-7 work to 3D datasets and a black-box regression model (IJHPCA).
- First Place, ACM Undergraduate Student Research Competition poster at SC '22.
- Contributed to Libpressio, an Argonne library for compression.

High Performance Computing Creative Inquiry, Clemson University — Jun 2021 – Dec 2022

- Best Poster (team) at the Indy Student Cluster Competition at SC '22.
- Student Cluster Competition at SC '21.
- Collaborated with Dell and Intel to configure a Pareto-optimal performance-per-watt cluster.
- Set up a distributed cluster with package managers, applications, and benchmarks.

Undergraduate Student Researcher, Clemson University, FTHPC — May 2021 – May 2022

- Lossy compression research with Argonne National Laboratory and Clemson University FTHPC.
- Analyzed statistical correlations within scientific 2D datasets in relation to compression performance.
- Publication at DRBSD-7, held during SC '21.

Peer-Reviewed Publications

[IJHPCA] Black-Box Statistical Prediction of Lossy Compression Ratios for Scientific Data
R. Underwood, J. Bessac, **D. Krasowska**, J. C. Calhoun, S. Di, F. Cappello. doi:10.1177/10943420231179417

[CLUSTER] A Lightweight, Effective Compressibility Estimation Method for Error-bounded Lossy Compression
A. Ganguli, R. Underwood, J. Bessac, **D. Krasowska**, J. C. Calhoun, S. Di, F. Cappello. doi:10.1109/CLUSTER52292.2023.00028

[DRBSD] Exploring Lossy Compressibility through Statistical Correlations of Scientific Datasets
D. Krasowska, J. Bessac, R. Underwood, J. C. Calhoun, S. Di, F. Cappello. doi:10.1109/DRBSD754563.2021.00011

Under Review

[ASPLOS] Easy, Yet Performant PIM Through Deferred Execution That Enables Vertical and Horizontal Fusion
D. Krasowska, A. Crotty, P. Dinda. Submitted to ASPLOS, 2026

[IPDPS] cuNumeric.jl: Automated Performant Multi-GPU Numerical Computing for Julia
D. Krasowska, E. Meitz, W. Lee, P. Dinda. Submitted to IPDPS, 2026

Technical Reports

[NU-CS] Village: From High Level Parallelism to High Performance
K. Nagaitsev, G. Dube, K. Hallsby, P. Liu, Q. Jiang, L. Myers, **D. Krasowska**, A. Butler, R. Xu, P. Dinda. Technical Report NU-CS-2025-33

Teaching & Mentoring

Teaching Assistant, CS 343: Operating Systems — Winter 2025
Zhehao Zhu, Master's Research Mentor — 2024 – 2025
Nader Rahhal, Undergraduate Research Mentor — 2025 – 2026

Awards

DOE Computational Science Graduate Fellowship, Department of Energy — 2023
First Place, ACM Undergraduate Student Research Competition, Supercomputing 2022 — 2022
Best Poster (Team), IndySCC22, Supercomputing 2022 — 2022